2. Basic Topology

Definition 2.1. *Let A and B be sets. If there exists a* 1-1 *mapping from A onto B, then we say that A and B can be put in* 1-1 **correspondence**, *that A and B have the same* **cardinal number (or cardinality)**, *or briefly A and B are* **equivalent**. *We write $A \sim B$ if two sets A and B are equivalent. One can easily check that the following three properties hold:*

(*1*) *It is reflexive : $A \sim A$*
(*2*) *It is symmetric : If $A \sim B$, then $B \sim A$*
(*3*) *It is transitive : If $A \sim B$ and $B \sim C$, then $A \sim C$.*

Exercise 2.2. *Prove the three properties of the relation $\sim$ above.*

Definition 2.3. *Let $\mathbb{N}$ be the natural number system, that is, $\mathbb{N}$ is the set of all positive integers. By J_n, we denote the set whose elements are the integers $1, 2, \ldots, n$. For any set A, we say:*

(*a*) *A is* **finite** *if $A \sim J_n$ for some $n \in \mathbb{N}$ or $A = \emptyset$.*
(*b*) *A is* **infinite** *if A is not finite.*
(*c*) *A is* **countable** *if $A \sim \mathbb{N}$.*
(*d*) *A is* **uncountable** *if A is neither finite nor countable*
(*e*) *A is* **at most countable** *if A is finite or countable.*

Exercise 2.4. *Let A and B be two finite sets. Assume that A and B contain the same number of elements. Show that $A \sim B$.*

Exercise 2.5. *Let A be an infinite set. Show that the number of elements of A is infinite.*

Exercise 2.6. *Let A be a finite set and B be a proper subset of A, that is, $B \subset A$ and $B \neq A$. Show that A and B cannot be equivalent.*

Remark 2.7. In many books, "countable" and "at most countable" are not distinguished. That is, usually a finite set is regarded as a countable set in many books. However, keep in mind that we do not define a finite set as a countable set in our lecture note (because Rudin did so).

Exercise 2.8. *Let A be a countable set. Show that A is infinite.*

Example 2.9. *By $\mathbb{Z}$, we denote the integer number system, that is, $\mathbb{Z}$ is the set of all integers. For any $n \in \mathbb{N}$, define*

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{n-1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

Then

$$f(1) = 0,\ f(2) = 1,\ f(3) = -1,\ f(4) = 2,\ f(5) = -2, \ldots.$$

Thus f is a 1-1 mapping from $\mathbb{N}$ onto $\mathbb{Z}$. Therefore, $\mathbb{Z}$ is countable since $\sim$ is symmetric.

Remark 2.10. By Exercise 2.6, a finite set cannot be equivalent to one of its proper subsets. However, Example 2.9 shows that it is possible for infinite sets. In other words, if a set A has a proper subset of A which is equivalent to itself A, then A is an infinite set.

Definition 2.11. *Any function f defined on $\mathbb{N}$ is called a* **sequence**. *If $f(n) = x_n$ for $n \in \mathbb{N}$, it is customary to denote the sequence f by the symbol $\{x_n\}$ or $x_1, x_2, x_3, \ldots$. The values of f, that is, the elements x_n are called the* **terms** *of the sequence. In particular, x_n is called the* **n-th term** *of the sequence. If there exists a set A such that $x_n \in A$ for all $n \in \mathbb{N}$, then $\{x_n\}$ is said to be a* **sequence** *in A, a* **sequence of elements** *of A, or A-valued sequence.*

Remark 2.12. Since every countable set is the range of a 1-1 function defined on $\mathbb{N}$, we may regard every countable set as the range of a sequence of distinct terms. In other words, if A is a countable set, we can put

$$A = \{x_n\}, \quad A = \{x_n : n \in \mathbb{N}\}, \quad \text{or} \quad A = \{x_1, x_2, \ldots\},$$

where $x_i \neq x_j$ whenever $i \neq j$.

Theorem 2.13. *Every infinite subset of a countable set is countable.*

Proof. Let A be a countable set and $E \subset A$. Assume that E is infinite. Arrange the elements x of A in a sequence $\{x_n\}$ of distinct elements. Let n_1 be the smallest positive integer such that $x_{n_1} \in E$. Inductively, for $k = 2, 3, 4, \ldots$, let n_k be the smallest integer grater than n_{k-1} such that $x_{n_k} \in E$. This is possible for all $k = 2, 3, 4, \ldots$ since E is an infinite set. By putting $f(k) = x_{n_k}$ for all $k \in \mathbb{N}$, we obtain a 1-1 correspondence between $\mathbb{N}$ and E. Therefore E is countable. $\square$

Remark 2.14. Obviously, Theorem 2.13 says that every subset of a countable set is at most countable.

Exercise 2.15. *Let E and A be sets such that $E \subset A$. Show that A is uncountable if E is uncountable.*

Definition 2.16. *Let A and Ω be sets. Assume that for each $\alpha \in A$, there exists a subset $E_\alpha \subset \Omega$. Then A is called an* **index set** *and the set whose elements are the sets E_α will be denoted by $\{E_\alpha\}$ or $\{E_\alpha : \alpha \in A\}$. A set of sets are usually called* **a collection of sets** *or* **a family of sets**. *The* **union** *of the sets E_α denoted by $\bigcup_{\alpha \in A} E_\alpha$ is defined to be the set such that $x \in \bigcup_{\alpha \in A} E_\alpha$ if and only if $x \in E_\alpha$ for some $\alpha \in A$. That is,*

$$\bigcup_{\alpha \in A} E_\alpha := \{x \in \Omega : x \in E_\alpha \text{ for some } \alpha \in A\}.$$

The **intersection** *of the sets E_α denoted by $\bigcap_{\alpha \in A} E_\alpha$ is defined to be the set such that $x \in \bigcap_{\alpha \in A} E_\alpha$ if and only if $x \in E_\alpha$ for all $\alpha \in A$. That is,*

$$\bigcap_{\alpha \in A} E_\alpha := \{x \in \Omega : x \in E_\alpha \text{ for all } \alpha \in A\}.$$

In particular, if $A = \{1, 2, \ldots, n\}$, we write

$$\bigcup_{m=1}^{n} E_m := \bigcup_{\alpha \in A} E_\alpha$$

and

$$\bigcap_{m=1}^{n} E_m := \bigcap_{\alpha \in A} E_\alpha.$$

Similarly, if $A = \mathbb{N}$, we denote

$$\bigcup_{m=1}^{\infty} E_m := \bigcup_{\alpha \in A} E_\alpha$$

and

$$\bigcap_{m=1}^{\infty} E_m := \bigcap_{\alpha \in A} E_\alpha.$$

For two sets A and B, if $A \cap B \neq \emptyset$, then we say that A and B are **intersect**. *On the other hand, if $A \cap B = \emptyset$, then A and B said to be* **disjoint**.

Theorem 2.17. *Let $\{E_n\}$ be a sequence of countable sets. Then*

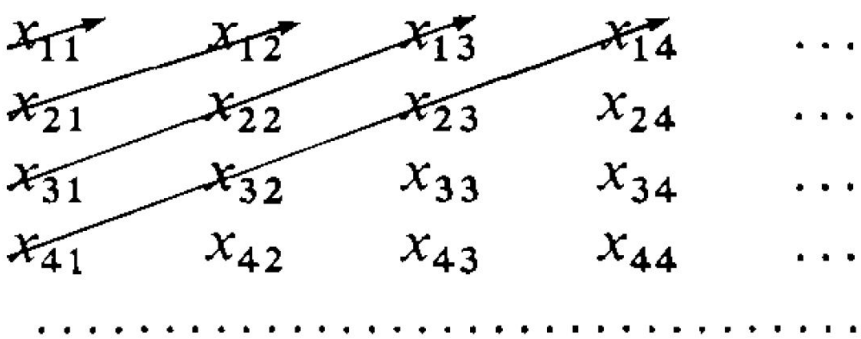
$$\bigcup_{n=1}^{\infty} E_n$$

is countable.

Proof. Since each E_n is a countable set, each E_n can be arranged in a sequence. Put

$$E_n = \{x_{nk}\}, \qquad (k = 1, 2, 3, \ldots).$$

We can count the all elements of $\bigcup_{n=1}^{\infty} E_n$ as follows:



As indicated by the arrows, these elements can be arranged in a sequence

$$x_{11}; x_{21}, x_{12}; x_{31}, x_{22}, x_{13}; x_{41}, x_{32}, x_{23}, x_{14}; \ldots. \tag{2.1}$$

If any two of the sets E_n have elements in common, these will appear more than once in (2.1). Hence there is a subset T of $\mathbb{N}$ such that

$$T \sim \bigcup_{n=1}^{\infty} E_n.$$

Moreover, since

$$E_1 \subset \bigcup_{n=1}^{\infty} E_n$$

and E_1 is an infinite set, $\bigcup_{n=1}^{\infty} E_n$ is an infinite set. Therefore by Theorem 2.13, $\bigcup_{n=1}^{\infty} E_n$ is countable. $\square$

Corollary 2.18. *Let A be a at most countable set. Assume that for each $\alpha \in A$, there exists a at most countable set B_α. Then*

$$\bigcup_{\alpha \in A} B_\alpha$$

is at most countable.

We left the proof of this corollary as an exercise.

Exercise 2.19. *Prove Corollary 2.18.*

Exercise 2.20. *Let A be a countable set. Assume that for each $\alpha \in A$, there exists a countable set E_α. Show that*

$$\bigcup_{\alpha \in A} E_\alpha$$

is a countable set.

Theorem 2.21. *Let A be a countable set and B_n be the set of all n-tuples $(a_1, \ldots, a_n)$ where $n \in \mathbb{N}$ and $a_k \in A$ for all $k = 1, \ldots, n$. Then B_n is countable.*

Proof. Obviously if $n = 1$, then B_1 is countable since $B_1 = A$. To prove that B_n is countable for all $n \in \mathbb{N}$, we use the mathematical induction. Suppose that B_{n-1} is countable for some $n \in \mathbb{N}$ with $n \geq 2$. The elements of B_n are of the form

$$(b, a) \qquad (b \in B_{n-1},\ a \in A).$$

For every fixed b,

$$\{(b, a) : a \in A\} \sim A.$$

Thus $\{(b, a) : a \in A\}$ is countable. Observe that

$$B_n = \bigcup_{b \in B_{n-1}} \{(b, a) : a \in A\}.$$

In other words, B_n is the countable union of the countable sets. Therefore, by Theorem 2.17, B_n is countable. □

Recall that $\mathbb{Q}$ denotes the set of all rational numbers.

Corollary 2.22. *$\mathbb{Q}$ is countable.*

Proof. Recall that $\mathbb{Z}$ is the set of all integer numbers and is countable (see Example 2.9). Then by Theorem 2.21, the Cartesian product $\mathbb{Z} \times \mathbb{Z}$ is countable. Note that every rational number is of the form $\frac{b}{a}$, where a, b are integers and $a \neq 0$. Therefore, there exists a subset $A \subset \mathbb{Z} \times \mathbb{Z}$ such that

$$A \sim \mathbb{Q}.$$

Since $\mathbb{Q}$ is an infinite set, A is infinite. Thus A is countable due to Theorem 2.13 and therefore $\mathbb{Q}$ is countable. □

Theorem 2.23. *Let A be the set of all sequences whose elements are the digits 0 and 1. Then A is uncountable.*

Proof. Note that the elements of A are sequences like $1, 0, 0, 1, 0, 1, 1, 1, \ldots$. We use the reduction to absurdity. Assume that A is countable. Then A can be arranged in a sequence $\{x_n\}$. In other words, for any element $\alpha \in A$, there exists $n \in \mathbb{N}$ such that $\alpha = x_n$. By the definition of A, each x_n is a form of $s_1, s_2, s_3, \ldots$, where $s_k = 0$ or $s_k = 1$ for all $k \in \mathbb{N}$. Thus we can put

$$x_n = s_{n1}, s_{n2}, s_{n3}, \ldots,$$

where $s_{nk} = 0$ or $s_{nk} = 1$ for all $n \in \mathbb{N}$ and $k \in \mathbb{N}$. Define the sequence $\tilde{x}$ as

$$\tilde{x} = 1 - s_{11}, 1 - s_{22}, 1 - s_{33}, 1 - s_{44}, \ldots.$$

Then obviously, for each $n \in \mathbb{N}$, the n-th term of $\tilde{x}$ differs from the n-th term of x_n since

$$1 - s_{nn} \neq s_{nn}.$$

Therefore, $\tilde{x} \neq x_n$ for all $n \in \mathbb{N}$. However, obviously, $\tilde{x} \in A$. It is contradiction. □

Remark 2.24. The method constructing $\tilde{x}$ in the proof of Theorem 2.23 is called **Cantor's diagonal process**.

Exercise 2.25. *Show that the real number system $\mathbb{R}$ is uncountable.*

For a function f, we use the notation $f : A \to B$ if the domain of f is A and the target space is B.

Definition 2.26. *A set X is said to be a* **metric space** *if and only if there exists a function $d : X \times X \to \mathbb{R}$ such that*

(*a*)
$$d(p, p) = 0 \qquad \forall p \in X$$

(*b*)
$$d(p, q) > 0 \quad \textit{if} \quad p \neq q$$

(*c*)
$$d(p, q) = d(q, p) \qquad \forall p, q \in X$$

(*d*) (*triangle inequality*)
$$d(p, q) \leq d(p, r) + d(r, q) \qquad \forall p, q, r, \in X.$$

Any function satisfying above four properties is called a **metric** *or* **distance function**. *An element of a metric space X is called a* **point**.

Sometime we say that (X, d) is a metric space if X is a metric space with the metric d.

Remark 2.27. Let (X, d) be a metric space. Then any subset of X is also an metric space with the metric d.

Example 2.28. *One of the most important examples of a metric space are the Euclidean spaces $\mathbb{R}^k$ (see Definition 1.32). The metric in $\mathbb{R}^k$ is defined by*

$$d(x, y) = |x - y| \qquad \forall x, y \in \mathbb{R}^k. \tag{2.2}$$

Due to Theorem 1.33, one can easily check that the function d defined in (2.2) *is indeed a metric, that is, d satisfies the four properties* (*a*), (*b*), (*c*), (*d*) *in Definition 2.26. If there is no information about the metric in $\mathbb{R}^k$, we assume that the given metric in $\mathbb{R}^k$ is* (2.2), *which is called the* **standard metric**.

Definition 2.29.

- *Let $a < b$. By the* **segment** (a, b)*, we mean the set of all real numbers x such that $a < x < b$. (a, b) is also called an* **open interval**. *By the* **interval** $[a, b]$*, we denote the set of all real numbers such that $a \leq x \leq b$. $[a, b]$ is also called a* **closed interval**. *Similarly, we define*

$$[a, b) := \{x \in \mathbb{R} : a \leq x < b\} \quad \text{and} \quad (a, b] := \{x \in \mathbb{R} : a < x \leq b\}.$$

- *If $a_i < b_i$ for $i = 1, \ldots, k$, the set of all points $x = (x_1, \ldots, x_k)$ in $\mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x_i \leq b_i$. is called a* **k-cell**. *That is, k-cell is the set $[a_1, b_1] \times \cdots \times [a_k, b_k]$. In particular, 1-cell is an closed interval and a 2-cell is a rectangle.*
- *If $x \in \mathbb{R}^k$ and $r > 0$, the* **open ball** *B with center at x and radius r is defined to be the set of all $y \in \mathbb{R}^k$ such that $|y - x| < r$. Similarly, the* **closed ball** *$\overline{B}$ with center at x and radius r is defined to be the set of all $y \in \mathbb{R}^k$ such that $|y - x| \leq r$. Sometime we write $B_r(x)$ and $\overline{B}_r(x)$ to emphasize the center x and the radius r.*
- *We call a set $E \subset \mathbb{R}^k$* **convex** *if*

$$\lambda x + (1 - \lambda)y \in E \qquad \forall x, y \in E \text{ and } 0 < \lambda < 1.$$

In particular, balls are convex. Indeed, by the triangle inequality, if $|y - x| < r$, $|z - x| < r$, and $0 < \lambda < 1$, then we have

$$\begin{aligned} |\lambda y + (1 - \lambda)z - x| &= |\lambda(y - x) + (1 - \lambda)(z - x)| \\ &\leq \lambda|y - x| + (1 - \lambda)|z - x| < \lambda r + (1 - \lambda)r = r. \end{aligned}$$

The same proof can apply to closed balls. Moreover one can easily check that k-cells are convex.

Exercise 2.30. *Show that k-cells are convex.*

Definition 2.31. *Let (X, d) be metric space. All points and sets mentioned below are understood to be elements and subsets of X.*

(*a*) *Let $r > 0$. A* **neighborhood** *of a point p is the set $N_r(p)$ consisting of all points q such that $d(p, q) < r$. That is,*

$$N_r(p) := \{q \in X : d(p, q) < r\}.$$

The number r is called the **radius** *of $N_r(p)$ and the point p is called the* **center** *of $N_r(p)$.*

(*b*) *A point p is a* **limit point** *of the set E if every neighborhood of p contains a point $q \neq p$ such that $q \in E$. That is, p is a limit point of E iff(:= if and only if) for all $r > 0$, there exists $q \in E$ such that $q \neq p$ and $q \in N_r(p)$.*

(*c*) *If $p \in E$ and p is not a limit point of E, then p is called an* **isolated point** *of E.*

(*d*) *E is* **closed** *if every limit point of E is a point of E.*

(*e*) *A point p is an* **interior point** *of E if there is a $N_r(p)$ (neighborhood of p) such that $N_r(p) \subset E$.*

(*f*) *E is* **open** *if every point of E is an interior point of E*

(*g*) *The* **complement** *of E (denoted by E^c) is the set of all points $p \in X$ such that $p \notin E$. That is,*

$$E^c := X - E := \{p \in X : p \notin E\}.$$

(*h*) *E is* **perfect** *if E is closed and every point of E is a limit point of E.*

(*i*) *E is* **bounded** *if there is a real number M and a point $q \in X$ such that $d(p,q) < M$ for all $p \in E$. That is, E is bounded iff there exists a neighborhood $N_M(q)$ such that $E \subset N_M(q)$.*
(*j*) *E is* **dense** *in X if every point of X is a limit point of E, or a point of E (or both). That is, E is dense in X iff for all $x \in X$ and $r > 0$, there exists a $q \in E$ such that $q \in N_r(x)$.*

Exercise 2.32. *Show that the rational number system $\mathbb{Q}$ is dense in the real number system $\mathbb{R}$.*

Theorem 2.33. *Let (X,d) be a metric space. Then every neighborhood in X is an open set.*

Proof. Let E be a neighborhood. Then there exist a point $p \in X$ and $r > 0$ such that $E = N_r(p)$. It is sufficient to show that any $q \in N_r(p)$ is an interior point. Let $q \in N_r(p)$. Then

$$d(p,q) < r$$

and thus $h := r - d(p,q) > 0$. Consider the neighborhood of q with the radius h, which is denoted by $N_h(q)$. Then for all $s \in N_h(q)$, by the triangle inequality in the metric space,

$$d(s,p) \leq d(s,q) + d(q,p) = d(s,q) + d(p,q) < h + d(p,q) = r - d(p,q) + d(p,q) = r.$$

Therefore

$$N_h(q) \subset N_r(p),$$

which implies that q is an interior point of $N_r(p)$. □

Theorem 2.34. *Let (X,d) be a metric space, E be a subset of X, and p be a limit point of E. Then every neighborhood of p contains infinitely many points of E.*

Proof. We use the reduction to absurdity. Suppose that the statement is not true. In other words, assume that there exists a neighborhood of p contains only finite many points of E and by $q_1, \ldots, q_n$ we denote these points which are distinct from p. Here $n \geq 1$ since p is a limit point and thus there exists at least one point which is different from p. Define

$$r := \min_{1 \leq m \leq n} d(p, q_m) := \text{ the minimum of } \{d(p,q_1), \ldots, d(p,q_m)\},$$

where min denotes the minimum. Since $d(p,q_i) > 0$ ($\because q_i \neq p$) for all $i = 1, \ldots, m$ and $\{d(p,q_i) : i = 1, \ldots, m\}$ is finite set, we have $r > 0$. Then the neighborhood $N_r(p)$ contains no point q of E such that $q \neq p$. Thus p is not a limit point of E, which is contradiction to the assumption. □

Theorem 2.35. *Let (X,d) be a nonempty metric space and E be a finite subset of X. Prove that*

(*i*) *E is bounded*
(*ii*) *E has no limit points*
(*iii*) *E is closed*

Proof. If $E = \emptyset$, then all statements are obvious and so we may assume that E is not an empty set. Set $n = |E|$, where $|E|$ denotes the number of elements of E. Then $n \in \mathbb{N}$. By $a_1, a_2, \ldots, a_n$, we denote the elements of E, i.e.

$$E = \{a_1, a_2, \ldots, a_n\}.$$

First we prove (i). Set

$$M = \max\{d(a_1, a_i) : i = 1, \ldots, n\} + 1,$$

where max denotes the maximum. Then $d(a_1, a_i) < M$ for all $i = 1, \ldots, n$. Therefore E is bounded.

Next we prove (ii). Assume that E has a limit point in X. We choose a limit point of E and denote it by p. By Theorem 2.34, for any $r > 0$, $N_r(p)$ contains infinitely many points of E. It is contradiction and thus (ii) is proved.

Finally we prove (iii). By (ii), the set of all limit points of E is empty set. Since $\emptyset \subset E$, E is closed. □

Example 2.36. *We consider a subset in $\mathbb{R}$. Let*

$$E := \left\{ \frac{1}{n} \in \mathbb{R} : n \in \mathbb{N} \right\}.$$

Then

(*i*) *E has only one limit point* 0

(*ii*) *E has no interior point of E*
(*iii*) *E is not closed*
(*iv*) *E is not open*
(*v*) *E is not perfect.*

Thus there exists a subset of $\mathbb{R}$ which is not open, closed, nor perfect.

Exercise 2.37. *Prove the statements (i)-(v) in Example 2.36.*

Example 2.38. *Set $E_1 := (a, b)$ and $E_2 := (a, b) \times \{0\}$.*

(*i*) *Recall that $d(x, y) := |x - y|$ for all $x, y \in \mathbb{R}$ is a metric in $\mathbb{R}$. By Theorem 2.33, E_1 is open in $\mathbb{R}$ since*

$$(a, b) = N_r(p),$$

where $p = \frac{a+b}{2}$ and $r = b - \frac{a+b}{2} = \frac{b-a}{2}$.

(*ii*) *Any point of E_2 is not an interior point of E_2 in $\mathbb{R}^2$.*

Therefore open interval is open in $\mathbb{R}$ but is not open in $\mathbb{R}^2$.

Lemma 2.39. *Let (X, d) be metric space, $E \subset X$, and x be a point in X. Assume that $x \notin E$ and x is not a limit point of E. Then there exists a $r > 0$ such that $N_r(x) \cap E = \emptyset$.*

Proof. If p is a limit point of E, then every neighborhood of p contains a point $q \neq p$ such that $q \in E$. Since x is not a limit point of E, not every neighborhood of x contains a point $q \neq x$ such that $q \in E$. Thus there exists a neighborhood of x which does not contain a point $q \neq x$ such that $q \in E$. In other words, since every neighborhood of x can be represented by $N_r(x)$ with $r > 0$, there exists a $r > 0$ such that $N_r(x) \cap E = \{x\}$ or $N_r(x) \cap E = \emptyset$. Since $x \notin E$ by the assumption, there exists a $r > 0$ such that $N_r(x) \cap E = \emptyset$. □

Theorem 2.40. *A set E is open if and only if E^c is closed.*

Proof. First we prove " the if part $\Leftarrow$". Assume that E^c is closed. It suffices to show that any $x \in E$ is an interior point of E. Let $x \in E$. Then $x \notin E^c$ and thus x is not a limit point of E^c since E^c is closed. Since x is not a limit point of E^c and $x \notin E^c$, by Lemma 2.39 there exists a $r > 0$ such that $N_r(x)$ such that $N_r(x) \cap E^c = \emptyset$. Since $N_r(x) \cap E^c = \emptyset$, $N_r(x) \subset E$.

Next we prove " the only if part $\Rightarrow$". Suppose that E is open. We have to show that any limit point x of E^c is contained in E^c. Let x be a limit point of E^c. Then every neighborhood of x contains a point $q \neq x$ such that $q \in E^c$. Thus there is no neighborhood of x such that $N_r(x) \subset E$ and thus x is not an interior point of E. Since E is open, $x \notin E$. Therefore $x \in E^c$. □

Corollary 2.41. *A set F is closed if and only if its complemnt is open.*

Proof. Due to Theorem 2.40, F^c is open if and only if $(F^c)^c$ is closed. Since $(F^c)^c = F$, the corollary is proved. □

Lemma 2.42 (Generalization of De Morgan's law)**.** *Let A be an index set and assume that for each $\alpha \in A$, there exists a set E_α, i. e. a collection $\{E_\alpha : \alpha \in A\}$ is given. Then*

(*a*)

$$\left(\bigcup_{\alpha \in A} E_\alpha\right)^c = \bigcap_{\alpha \in A} E_\alpha^c.$$

(*b*)

$$\left(\bigcap_{\alpha \in A} E_\alpha\right)^c = \bigcup_{\alpha \in A} E_\alpha^c.$$

Proof. We only prove (a) because of the similarity of the proof. First we prove

$$\left(\bigcup_{\alpha \in A} E_\alpha\right)^c \subset \bigcap_{\alpha \in A} E_\alpha^c.$$

Let $x \in \left(\bigcup_{\alpha \in A} E_\alpha\right)^c$. Then

$$x \notin \bigcup_{\alpha \in A} E_\alpha$$

and thus $x \notin E_\alpha$ for all $\alpha \in A$. By the definition of E_α^c, $x \in E_\alpha^c$ for all $\alpha \in A$. Therefore by the definition of $\bigcap_{\alpha \in A} E_\alpha^c$, we have

$$x \in \bigcap_{\alpha \in A} E_\alpha^c.$$

Next we prove

$$\left(\bigcup_{\alpha \in A} E_\alpha\right)^c \supset \bigcap_{\alpha \in A} E_\alpha^c.$$

Let $x \in \bigcap_{\alpha\in A} E_\alpha^c$. Then $x \in E_\alpha^c$ for all $\alpha \in A$ and thus $x \notin E_\alpha$ for all $\alpha \in A$. Therefore $x \notin \bigcup_{\alpha\in A} E_\alpha$ and finally we have

$$x \in \left(\bigcup_{\alpha\in A} E_\alpha\right)^c.$$

The lemma is proved. □

Theorem 2.43. (*a*) *Let A be an index set and assume that for each $\alpha \in A$, there exists an open set G_α. Then $\bigcup_{\alpha\in A} G_\alpha$ is open.*

(*b*) *Let A be an index set and assume that for each $\alpha \in A$, there exists a closed set F_α. Then $\bigcap_{\alpha\in A} F_\alpha$ is closed.*

(*c*) *Let $n \in \mathbb{N}$ and assume that there exist open sets $G_1, \cdots, G_n$. Then $\cap_{i=1}^n G_i$ is open.*

(*d*) *Let $n \in \mathbb{N}$ and assume that there exist closed sets $F_1, \cdots, F_n$. Then $\cup_{i=1}^n F_i$ is closed.*

Proof. (a) It suffices to show that for any $x \in \bigcup_{\alpha\in A} G_\alpha$, x is an interior point of $\bigcup_{\alpha\in A} G_\alpha$. Let $x \in \bigcup_{\alpha\in A} G_\alpha$. Then by the definition of the union, there exists a $\alpha_0 \in A$ such that $x \in G_{\alpha_0}$. Since G_{α_0} is open, x is an interior point of G_{α_0}. Thus there exists a $r > 0$ such that $N_r(x) \subset G_{\alpha_0}$. Since $G_{\alpha_0} \subset \bigcup_{\alpha\in A} G_\alpha$, we have

$$N_r(x) \subset G_{\alpha_0} \subset \bigcup_{\alpha\in A} G_\alpha.$$

Therefore x is an interior point of $\bigcup_{\alpha\in A} G_\alpha$ and this proves (*a*).

(b) Due to Corollary 2.41, it suffices to show that $\left(\bigcap_{\alpha\in A} F_\alpha\right)^c$ is open. By Theorem 2.40, F_α^c is open for all $\alpha \in A$. Thus by (*a*), $\bigcup_{\alpha\in A} F_\alpha^c$ is also open. Moreover, due to Lemma 2.42,

$$\left(\bigcap_{\alpha\in A} F_\alpha\right)^c = \bigcup_{\alpha\in A} F_\alpha^c.$$

Therefore $\left(\bigcap_{\alpha\in A} F_\alpha\right)^c$ is open.

(c) It suffices to show that any point $x \in \cap_{i=1}^n G_i$ is an interior point. Let $x \in \cap_{i=1}^n G_i$. Then $x \in G_i$ for all $i = 1, \ldots, n$. Since each G_i is open, x is an interior point and thus for each $i = 1, \ldots, d$, there exists $r_i > 0$ such that $B_{r_i}(x) \subset G_i$. Put

$$r = \min\{r_1, \ldots, r_n\}.$$

Then obviously $r > 0$ and

$$N_r(x) \subset N_{r_i}(x) \qquad \forall i = 1, \ldots, n.$$

Therefore we have

$$N_r(x) \subset \cap_{i=1}^n G_i$$

and this shows that x is an interior point.

(d) Since the proof of (*d*) is similar to that of (*b*), we skip the details.

□

Exercise 2.44. *Prove* (*d*) *in Theorem 2.43.*

Theorem 2.45. *Disprove that for a sequence of opens sets $G_1, G_2, \cdots, \bigcap_{n=1}^\infty G_n$ is open.*

Proof. It suffices to find a counter example. That is, we need to find a sequence of open sets G_n such that $\bigcap_{n=1}^{\infty} G_n$ is not open. Let

$$G_n := \left(-\frac{1}{n}, \frac{1}{n}\right) \qquad (n = 1, 2, \cdots).$$

Then each G_n is open. However,

$$\bigcap_{n=1}^{\infty} G_n = \{0\}.$$

□

Exercise 2.46. *Prove or disprove that for a sequence of closed sets F_1, F_2, $\cdots$, $\bigcup_{n=1}^{\infty} F_n$ is closed or open.*

Definition 2.47. *Let X be a metric space.*

(*a*) *By E', we denote the set of all limit points of E in X.*
(*b*) *The* **closure** *of E is denoted by $\overline{E}$ and is defined as $\overline{E} = E \cup E'$.*

Lemma 2.48. *Let X be a metric space, $E \subset X$, and $N_r(x)$ be a neighborhood in X. Assume that $E \cap N_r(x) = \emptyset$. Then $\overline{E} \cap N_r(x) = \emptyset$.*

Proof. It suffices to show that $E' \cap N_r(x) = \emptyset$, i.e. there is no limit point of E in $N_r(x)$. We use the reduction to absurdity. Assume that there is a limit point p of E in $N_r(x)$. Then $d(p, x) < r$ and define $h := r - d(p, x) > 0$. Following the proof of Theorem 2.33, we can check that

$$N_h(p) \subset N_r(x).$$

However, since p is a limit point of E, there exists a $q \in E$ such that $q \neq p$ and $q \in N_h(p)$. Thus there exists a $q \in E \cap N_h(p) \subset E \cap N_r(x)$. It is contradiction to the assumption that $E \cap N_r(x) = \emptyset$. □

By using the above lemma, we can enhance the result in Lemma 2.39.

Corollary 2.49. *Let (X, d) be metric space, $E \subset X$, and x be a point in X. Assume that $x \notin E$ and x is not a limit point of E. Then there exists a $r > 0$ such that $N_r(x) \cap \overline{E} = \emptyset$.*

Exercise 2.50. *Prove Corollary 2.49.*

Theorem 2.51. *Let X be a metric space and $E \subset X$.*

(*a*) *$\overline{E}$ is closed*
(*b*) *$E = \overline{E}$ if and only if E is closed.*
(*c*) *For any closed set $F \subset X$ such that $E \subset F$, we have $\overline{E} \subset F$. In this sense, we say that the closure $\overline{E}$ is the smallest closed set containing E.*

Proof. (a) Due to Corollary 2.41, it suffices to show that $\overline{E}^c$ is open, i.e. for any $x \in \overline{E}^c$, x is an interior point of $\overline{E}^c$. Let $x \in \overline{E}^c$. Then $x \notin \overline{E}$. Thus $x \notin E$ and $x \notin E'$. Therefore by Corollary 2.49, there exists a $r > 0$ such that $N_r(x) \cap \overline{E} = \emptyset$, which implies $N_r(x) \subset \overline{E}^c$ and x is an interior point of $\overline{E}^c$

(b) First we prove the only if part (⇒). If $E = \overline{E}$, then by (*a*) E is closed. Next we prove the if part (⇐). Since $\overline{E} = E \cup E'$, obviously $E \subset \overline{E}$. Moreover, if E is closed, then $E' \subset E$. Therefore, if E is closed, then we have $\overline{E} = E$.

(c) It suffices to show that $E' \subset F$ since $E \subset F$. Observe that if $E \subset F$, then any limit point of E is also a limit point of F, i.e. $E' \subset F'$. Moreover since F is closed, $F' \subset F$. Therefore, we have

$$E' \subset F' \subset F.$$

□

Next we consider a subset of a metric space as an another metric space.

Remark 2.52. Let (X, d) be a metric space and $Y \subset X$. Then as mentioned in Remark 2.27, (Y, d) is a metric space. Then we can consider neighborhoods and open sets in the metric space Y with the metric d. Recall that for $r > 0$ and $p \in Y$, we define the neighborhood of p with the radius r as

$$N_r(p) := \{q \in Y : d(p, q) < r\}.$$

We use the notation $N_r^Y(p)$ instead of $N_r(p)$ to emphasize the related metric space. Since $N_r^Y(p) \subset Y \subset X$, it is obvious that $N_r^Y(p)$ is a subset of X. Thus it is natural to ask that whether $N_r^Y(p)$ is a neighborhood in X or not. It is proved that any neighborhood is an open set in Theorem 2.33. Therefore we can also ask whether $N_r^Y(p)$ is an open set in X or not.

Definition 2.53. *Let (X, d) be a metric space and $Y \subset X$. Then (Y, d) is a metric space and we can consider open sets in Y. To distinguish between open sets in Y and open sets in X, we say that a set E is open* **relative to** *Y if E is an open set in Y, i.e. for any $y \in E$, there exists a neighborhood $N_r^Y(y)$ such that $N_r^Y(y) \subset E$.*

Exercise 2.54. *Prove or disprove that*

(*a*) *$N_r^Y(p)$ is a neighborhood in X.*
(*b*) *$N_r^Y(p)$ is an open set in X.*

Theorem 2.55. *Let (X, d) be a metric space and $Y \subset X$. Then $E \subset Y$ is open relative to Y if and only if there exists an open set G relative to X such that $E = Y \cap G$.*

Proof. First we prove the only if part ($\Rightarrow$). Assume that $E \subset Y$ and E is open relative to Y. Then for any $y \in E$, there exists a $r_y > 0$ such that $N_{r_y}^Y(y) \subset E$. Define

$$G := \bigcup_{y \in E} N_{r_y}^X(y),$$

where $N_{r_y}^X(y) := \{q \in X : d(y, q) < r_y\}$. Then by Theorem 2.43(a), G is open in X Obviously, $E \subset G \cap Y$. Moreover, since

$$N_{r_y}^X(y) \cap Y = N_{r_y}^Y(y),$$

we have

$$G \cap Y = \bigcup_{y \in E} \left(N_{r_y}^X(y) \cap Y \right) = \bigcup_{y \in E} N_{r_y}^Y(y) \subset E.$$

Next we prove the if part ($\Leftarrow$). Assume that G is open in X and $E = G \cap Y$. Since G is open relative to X, for each $x \in G$, there exists a $r_x > 0$ such that $N_{r_x}^X(x) \subset G$. Let $x \in E$. Then since

$$N_{r_x}^X(x) \cap Y = N_{r_x}^Y(x),$$

we have

$$N_{r_x}^Y(x) \subset G \cap Y = E.$$

Therefore x is an interior point of E in Y and it implies E is open relative to Y. □

Exercise 2.56. *Let (X, d) be a metric space and $Y \subset X$. We say that E is closed relative to Y if E is a closed set in the metric space (Y, d). Compare the closed sets relative to X and the closed sets relative to Y.*

Definition 2.57. *Let X be a metric space and $E \subset X$. An* **open cover** *of E is a collection of open subsets of X whose union covers E. In other words, a collection of open sets $\{G_\alpha : \alpha \in A\}$ such that*

$$E \subset \bigcup_{\alpha \in A} G_\alpha$$

is an open cover of E. Let $\{G_\alpha : \alpha \in A\}$ be an open cover of E. Assume that there exists a $A_0 \subset A$ such that $\{G_\alpha : \alpha \in A_0\}$ such that

$$E \subset \bigcup_{\alpha \in A_0} G_\alpha.$$

Then the subcollection $\{G_\alpha : \alpha \in A_0\}$ of $\{G_\alpha : \alpha \in A\}$ is called (open) **subcover.** *Moreover, this subcover is called* **finite subcover** *if A_0 is finite.*

Definition 2.58. *Let X be a metric space and $K \subset X$. K is said to be* **compact** *iff every open cover of K has a finite subcover. In other words, K is compact iff for any open cover $\{G_\alpha : \alpha \in A\}$ of K, there exists a finite set $A_0 \subset A$ such that*

$$K \subset \bigcup_{\alpha \in A_0} G_\alpha.$$

Since A_0 is a finite set, we can put $A_0 = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ for some $n \in \mathbb{N}$. Thus K is compact iff for any open cover $\{G_\alpha : \alpha \in A\}$ of K, there exist α_1, α_2, …, α_n such that

$$K \subset G_{\alpha_1} \cup G_{\alpha_2} \cup \cdots \cup G_{\alpha_n}$$

Definition 2.59. *Let X be a metric space and $E \subset Y \subset X$.*

- *We call an open cover of E in Y* **an open cover of E relative to Y**, *i.e. an open cover of E relative to Y is a collection of open sets relative to Y whose union covers E. In other words, a collection of open sets relative to Y $\{G_\alpha^Y : \alpha \in A\}$ such that*

  $$E \subset \bigcup_{\alpha \in A} G_\alpha^Y$$

  is an open cover of E relative to Y. Due to Theorem 2.55, G_α^Y is open relative to Y iff there exists an open set G_α in X such that $G_\alpha^Y = G_\alpha \cap Y$. Thus any open cover of E relative to Y is given by

  $$\{G_\alpha \cap Y : \alpha \in A\},$$

  where A is an index set and each G_α is open relative to X.

- *A set $K \subset Y$ is* **compact relative to** *Y iff for any open cover of K relative to Y, there exist a finite subcover of K relative to Y. In other words, K is compact relative to Y iff for any open cover of K relative to Y $\{G_\alpha^Y : \alpha \in A\}$, there exist α_1, α_2, …, α_n such that*
$$K \subset G_{\alpha_1}^Y \cup G_{\alpha_2}^Y \cup \cdots \cup G_{\alpha_n}^Y.$$

Remark 2.60. Let X be a metric space and $K \subset Y \subset X$. If we say that K is compact without mentioning any relativity, then conventionally it means that X is compact relative to the whole space X. Customarily, the similar rule applies to open sets, closed sets, and open covers.

Exercise 2.61. *Define a finite subcover of E relative to Y.*

Theorem 2.62. *Let X be a metric space and $K \subset Y \subset X$. Then K is compact relative to X if and only if K is compact relative to Y.*

Proof. Prove the only if part ($\Rightarrow$) first. Assume that K is compact relative to X. Let $\{G_\alpha^Y : \alpha \in A\}$ be an open cover of K relative to Y. It suffices to show that there exist $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that
$$K \subset G_{\alpha_1}^Y \cup G_{\alpha_2}^Y \cup \cdots \cup G_{\alpha_n}^Y.$$
Due to Theorem 2.55, for each $\alpha \in A$, there exist an open set G_α relative to X such that
$$G_\alpha^Y = G_\alpha \cap Y.$$
Since $\{G_\alpha^Y : \alpha \in A\}$ is an open cover of K relative to Y, we have
$$K \subset \bigcup_{\alpha \in A} G_\alpha^Y \subset \bigcup_{\alpha \in A} G_\alpha.$$
Thus $\{G_\alpha : \alpha \in A\}$ is an open cover of K relative to X. By the assumption that K is compact relative to X, there exist $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that
$$K \subset G_{\alpha_1} \cup G_{\alpha_2} \cup \cdots \cup G_{\alpha_n}.$$
Therefore, since $K \subset Y$, we have
$$\begin{aligned} K \subset (G_{\alpha_1} \cup G_{\alpha_2} \cup \cdots \cup G_{\alpha_n}) \bigcap Y &= (G_{\alpha_1} \cap Y) \cup (G_{\alpha_2} \cap Y) \cup \cdots \cup (G_{\alpha_n} \cap Y) \\ &= G_{\alpha_1}^Y \cup G_{\alpha_2}^Y \cup \cdots \cup G_{\alpha_n}^Y. \end{aligned}$$
Next we prove the if part ($\Leftarrow$). Assume that K is compact relative to Y. Let $\{G_\alpha : \alpha \in A\}$ be an open cover of K relative to X. Then $\{G_\alpha \cap Y : \alpha \in A\}$ is an open cover of K relative to Y. Since K is compact relative to Y, there exists $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that
$$K \subset (G_{\alpha_1} \cap Y) \cup (G_{\alpha_2} \cap Y) \cup \cdots \cup (G_{\alpha_n} \cap Y).$$
Since obviously $G_{\alpha_i} \cap Y \subset G_{\alpha_i}$ for all $i = 1, \ldots, n$, we have
$$K \subset G_{\alpha_1} \cup G_{\alpha_2} \cup \cdots \cup G_{\alpha_n}.$$
□

Theorem 2.63. *Let X be a metric space and $K \subset X$. Assume that K is compact. Then K is closed. In other words, compact subsets of metric spaces are closed.*

Proof. It suffices to show that K^c is open. Let $p \in K^c$. It is sufficient to show that p is an interior point of K^c. Then for any $q \in K$, $d(p,q) > 0$ since $p \neq q$. For $q \in K$, consider $N_{\frac{1}{2}d(p,q)}(p)$ and $N_{\frac{1}{2}d(p,q)}(q)$. Then

$$N_{\frac{1}{2}d(p,q)}(p) \cap N_{\frac{1}{2}d(p,q)}(q) = \emptyset. \tag{2.3}$$

Thus obviously

$$K \subset \bigcup_{q \in K} N_{\frac{1}{2}d(p,q)}(q).$$

and thus the collection

$$\{N_{\frac{1}{2}d(p,q)}(q) : q \in K\}$$

is an open cover of K. Therefore, since K is compact, there exist $q_1, \ldots, q_n$ such that

$$K \subset N_{\frac{1}{2}d(p,q_1)}(q_1) \cup \ldots \cup N_{\frac{1}{2}d(p,q_n)}(q_n)$$

Next set

$$V_p := N_{\frac{1}{2}d(p,q_1)}(p) \cap \ldots \cap N_{\frac{1}{2}d(p,q_n)}(p)$$

Then V_p is open and due to (2.3),

$$V_p \cap \left(N_{\frac{1}{2}d(p,q_1)}(q_1) \cup \ldots \cup N_{\frac{1}{2}d(p,q_n)}(q_n)\right) = \emptyset.$$

Thus since

$$K \subset N_{\frac{1}{2}d(p,q_1)}(q_1) \cup \ldots \cup N_{\frac{1}{2}d(p,q_n)}(q_n),$$

we have

$$V_p \cap K = \emptyset$$

and it implies $V_p \subset K^c$. Moreover, since V_p is open and $p \in V_p$, there exists $N_r(p)$ such that $N_r(p) \subset V_p$. Therefore p is an interior point of K^c since $V_p \subset K^c$. □

Exercise 2.64. *Let X be a metric space, $K \subset X$, $p \in K^c$, and $q \in K$. Show that for all $0 < r \leq \frac{1}{2}d(p,q)$, two neighborhoods $N_r(p)$ are $N_r(q)$ are disjoint.*

Theorem 2.65. *Let X be a metric space and $F \subset K \subset X$. Assume that F is closed and K is compact. Then F is compact. In other words, closed subsets of compact sets are compact.*

Proof. Let $\{V_\alpha : \alpha \in A\}$ be an open cover of F. Then

$$F \subset \bigcup_{\alpha \in A} V_\alpha.$$

Since F is closed, F^c is open. Since $K \subset X = F \cup F^c$, we have

$$K \subset F \cup F^c \subset \left(\bigcup_{\alpha \in A} V_\alpha\right) \cup F^c.$$

Thus the collection

$$\{V_\alpha : \alpha \in A\} \cup \{F^c\}$$

is an open cover of K. Since K is compact, there exist $\alpha_1, \ldots, \alpha_n$ such that

$$K \subset V_{\alpha_1} \cup \cdots \cup V_{\alpha_n} \cup F^c.$$

Since $F \subset K$ and $F \cap F^c = \emptyset$, we have

$$F \subset V_{\alpha_1} \cup \cdots \cup V_{\alpha_n}.$$

In other words, $\{V_{\alpha_1}, \dots, V_{\alpha_n}\}$ is a finite open cover of F. Therefore F is compact. $\square$

Corollary 2.66. *If F is closed and K is compact, then $F \cap K$ is compact.*

Proof. By Theorem 2.63, K is closed. Thus by Theorem 2.43(b), $F \cap K$ is closed. Since $F \cap K \subset K$, $F \cap K$ is a closed subset of K. Therefore by Theorem 2.65, $F \cap K$ is compact. $\square$

Theorem 2.67. *Let $\{K_\alpha : \alpha \in A\}$ be a collection of compact subsets of a metric space X. Assume that the intersection of every (non-empty) finite subcollection of $\{K_\alpha : \alpha \in A\}$ is nonmepty, i.e. for any (non-empty) finite $B \subset A$,*

$$\bigcap_{\alpha \in B} K_\alpha \neq \emptyset.$$

Then

$$\bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

Proof. We use the reduction to absurdity. Assume that

$$\bigcap_{\alpha \in A} K_\alpha = \emptyset. \tag{2.4}$$

For each K_α, put $G_\alpha := K_\alpha^c$. Fix $K_{\alpha_0} \in \{K_\alpha : \alpha \in A\}$. Then due to (2.4) and the generalization of De Morgan's law,

$$K_{\alpha_0} \subset \left(\bigcap_{\alpha \in A - \{\alpha_0\}} K_\alpha \right)^c = \bigcup_{\alpha \in A - \{\alpha_0\}} G_\alpha.$$

Thus

$$\{G_\alpha : \alpha \in A - \{\alpha_0\}\}$$

is an open cover of K_{α_0}. Since K_{α_0} is compact, there exist a finite set $B = \{\alpha_1, \dots, \alpha_n\}$ such that

$$K_{\alpha_0} \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}.$$

By De Moregan's law,

$$G_{\alpha_1} \cup \cdots \cup G_{\alpha_n} = (K_{\alpha_1} \cap \cdots \cap K_{\alpha_n})^c.$$

Therefore, we have

$$K_{\alpha_0} \cap K_{\alpha_1} \cap \cdots \cap K_{\alpha_n} = \emptyset$$

and it is contradiction to our assumption that every finite subcollection of $\{K_\alpha : \alpha \in A\}$ is nonmepty. $\square$

Exercise 2.68. *Let $\{K_\alpha : \alpha \in A\}$ be a collection of closed subsets of a metric space X. Assume that the intersection of every finite subcollection of $\{K_\alpha : \alpha \in A\}$ is nonmepty, i.e. for any finite $B \subset A$,*

$$\bigcap_{\alpha \in B} K_\alpha \neq \emptyset.$$

Prove or disprove that

$$\bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

Exercise 2.69. *Let $\{K_\alpha : \alpha \in A\}$ be a collection of open subsets of a metric space X. Assume that the intersection of every finite subcollection of $\{K_\alpha : \alpha \in A\}$ is nonmepty, i.e. for any finite $B \subset A$,*

$$\bigcap_{\alpha \in B} K_\alpha \neq \emptyset.$$

Prove or disprove that

$$\bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

Corollary 2.70. *Let K_1, K_2, ... be a sequence of nonempty compact sets such that $K_n \supset K_{n+1}$ for all $n \in \mathbb{N}$. Then*

$$\bigcap_{n=1}^{\infty} K_n \neq \emptyset.$$

Exercise 2.71. *Prove Corollary 2.70.*

Theorem 2.72. *Let E be an infinite subset of a compact set K. Then E has a limit point in K.*

Proof. We use the reduction to absurdity. Assume that there is no limit point of E in K. Then for each $q \in K$, there exists a neighborhood $N_{r_q}(q)$ such that $N_{r_q}(q) \cap E = \emptyset$ or $N_{r_q}(q) \cap E = \{q\}$. Obviously,

$$K \subset \bigcup_{q \in K} N_{r_q}(q).$$

Thus the collection $\{N_{r_q}(q) : q \in K\}$ is an open cover of K. Since K is compact, there exists $q_1, \ldots, q_n \in K$ such that

$$K \subset N_{r_{q_1}}(q_1) \cup \cdots \cup N_{r_{q_n}}(q_n).$$

Since $E \subset K$,

$$E \subset K \cap E \subset \left(N_{r_{q_1}}(q_1) \cup \cdots \cup N_{r_{q_n}}(q_n)\right) \cap E.$$

Finally, since $N_{r_q}(q) \cap E = \emptyset$ or $N_{r_q}(q) \cap E = \{q\}$, we have

$$E \subset \{q_1, \ldots, q_n\}$$

and it is contradiction to the assumption that E is infinite. □

Theorem 2.73. *Let I_1, I_2, ... be a sequence of closed intervals in $\mathbb{R}$. Assume that $I_n \supset I_{n+1}$ for all $n \in \mathbb{N}$. Then*

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Proof. Since each I_n is a closed interval, there exist $a_n \leq b_n$ such that

$$I_n = [a_n, b_n].$$

Define

$$E = \{a_n : n \in \mathbb{N}\}.$$

Due to the assumption that $I_n \supset I_{n+1}$ for all $n \in \mathbb{N}$, for all positive integers n, m we have

$$[a_{n+m}, b_{n+m}] \subset [a_n, b_n]$$

and thus

$$a_n \leq a_{m+n} \leq b_{m+n} \leq b_n.$$

In particular,

$$a_n \leq b_m$$

for all $n, m \in \mathbb{N}$. Hence for all $m \in \mathbb{N}$, b_m is an upper bound of E and by the least upper bound property of $\mathbb{R}$, there exists $\sup E$. Since each $a_n \in E$, obviously

$$a_n \leq \sup E \qquad \forall n \in \mathbb{N}$$

Moreover, since each b_n is an upper bound of E, we have

$$\sup E \leq b_n \qquad \forall n \in \mathbb{N}$$

Therefore, we have

$$a_n \leq \sup E \leq b_n \qquad \forall n \in \mathbb{N}$$

and it implies

$$\sup E \in [a_n, b_n] = I_n \qquad \forall n \in \mathbb{N}.$$

and

$$\sup E \in \bigcap_{n=1}^{\infty} I_n.$$

□

Exercise 2.74. *Let $\{[a_n, b_n] : n \in \mathbb{N}\}$ be a sequence of closed intervals Assume that*

$$a_n \leq b_m$$

for all $n, m \in \mathbb{N}$ and define $E = \{b_n : n \in \mathbb{N}\}$.

(*a*) *Show that*

$$\inf E \in \bigcap_{n=1}^{\infty} I_n.$$

(*b*) *Prove or disprove that*

$$\inf E = \sup\{a_n : n \in \mathbb{N}\}.$$

Theorem 2.75. *Let $k \in \mathbb{N}$ and I_1, I_2, ... be a sequence of k-cells. Assume that*

$$I_n \supset I_{n+1} \qquad \forall n \in \mathbb{N}.$$

Then

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Proof. Since each I_n is a k-cell, there exist real numbers $a_{n,1} \leq b_{n,1}$, $a_{n,2} \leq b_{n,2}$, $\cdots$, $a_{n,k} \leq b_{n,k}$ such that

$$I_n = [a_{n,1}, b_{n,1}] \times [a_{n,2}, b_{n,2}] \times \cdots \times [a_{n,k}, b_{n,k}].$$

For each $n \in \mathbb{N}$ and $j = 1, 2, \ldots, k$, we define

$$I_{n,j} := [a_{n,j}, b_{n,j}].$$

Then

$$I_n = I_{n,1} \times I_{n,2} \times \cdots \times I_{n,k}. \tag{2.5}$$

Moreover, due to the assumption

$$I_n \supset I_{n+1} \qquad \forall n \in \mathbb{N},$$

for each $j = 1, 2, \ldots, k$, we have

$$I_{n,j} \supset I_{n+1,j} \qquad \forall n \in \mathbb{N}.$$

Thus by Theorem 2.73, for each $j = 1, \ldots, k$,

$$\bigcap_{n=1}^{\infty} I_{n,j} \neq \emptyset$$

and we can choose a $x_j^* \in \bigcap_{n=1}^{\infty} I_{n,j}$. Set

$$x^* = (x_1^*, x_2^*, \ldots, x_k^*).$$

Then finally by (2.5) we have

$$x^* \in \bigcap_{n=1}^{\infty} I_n$$

and it implies that

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

□

Theorem 2.76. *Every k-cell is compact.*

Proof. Let I be a k-cell. Then by the definition of the k-cell, there exist $a_j \leq b_j$ $(j = 1, \ldots, k)$ such that

$$I = [a_1, b_1] \times \cdots \times [a_k, b_k].$$

Put

$$\delta = \left(\sum_{j=1}^{k} (b_j - a_j)^2 \right)^{1/2}.$$

Then obviously

$$|x - y| \leq \delta \qquad \forall x, y, \in I.$$

We use the reduction to absurdity. Assume that I is not compact. Then there exists an open cover $\{G_\alpha : \alpha \in A\}$ of I which does not have a finite subcover of I. For $j = 1, \ldots, k$, put

$$c_j = \frac{a_j + b_j}{2},$$

$$I_{1,0,j} := [a_j, c_j], \quad \text{and} \quad I_{1,1,j} := [c_j, b_j].$$

Then

$$[a_j, b_j] = [a_j, c_j] \cup [c_j, b_j] = I_{1,0,j} \cup I_{1,1,j}.$$

For $i_1, i_2, \ldots, i_k \in \{0, 1\}$, define

$$I_{1,i_1,i_2,\ldots,i_k} := I_{1,i_1,1} \times I_{1,i_2,2} \times \cdots \times I_{1,i_k,k}$$

Then

$$I = \bigcup_{i_1,i_2,\ldots,i_k \in \{0,1\}} I_{1,i_1,1} \times I_{1,i_2,2} \times \cdots \times I_{1,i_k,k}.$$

Since the open cover $\{G_\alpha : \alpha \in A\}$ is an open cover of I, $\{G_\alpha : \alpha \in A\}$ is also an open cover of

$$I_{1,i_1,1} \times I_{1,i_2,2} \times \cdots \times I_{1,i_k,k}$$

for all $i_1, i_2, \ldots, i_k \in \{0,1\}$. Since $\{G_\alpha : \alpha \in A\}$ does not have a finite subcover of I, there exist $\tilde{i}_1, \tilde{i}_2, \ldots, \tilde{i}_k \in \{0,1\}$ such that the open cover $\{G_\alpha : \alpha \in A\}$ of $I_{1,\tilde{i}_1,1} \times I_{1,\tilde{i}_2,2} \times \cdots \times I_{1,\tilde{i}_k,k}$ does not have a finite subcover of $I_{1,\tilde{i}_1,1} \times I_{1,\tilde{i}_2,2} \times \cdots \times I_{1,\tilde{i}_k,k}$. Put

$$I_1 = I_{1,\tilde{i}_1,1} \times I_{1,\tilde{i}_2,2} \times \cdots \times I_{1,\tilde{i}_k,k}.$$

Then

$$I \supset I_1,$$

I_1 is not covered by any finite subcollection of $\{G_\alpha : \alpha \in A\}$, and

$$|x - y| \leq \frac{\delta}{2} \qquad \forall x, y \in I_1.$$

Similarly we can subdivide I_1 into 2^k k-cells. Then there exist an k-cell I_2 such that

$$I \supset I_1 \supset I_2,$$

I_2 is not covered by any finite subcollection of $\{G_\alpha : \alpha \in A\}$, and

$$|x - y| \leq \frac{\delta}{2^2} \qquad \forall x, y \in I_2.$$

Inductively, we can find I_1, I_2, I_3, ... such that

$$I \supset I_1 \supset I_2 \supset I_3 \supset I_4 \supset \cdots$$

I_n is not covered by any finite subcollection of $\{G_\alpha : \alpha \in A\}$, and

$$|x - y| \leq \frac{\delta}{2^n} \qquad \forall x, y \in I_n \tag{2.6}$$

for all $n \in \mathbb{N}$. By Theorem 2.75, there exists $x^* \in \cap_{n=1}^{\infty} I_n$. Thus

$$x^* \in I_n \qquad \forall n \in \mathbb{N}.$$

Since $\{G_\alpha : \alpha \in A\}$ is an open cover of I, there exists G_α such that

$$x^* \in G_\alpha.$$

Since G_α is an open set, there exists a neighborhood $N_r(x^*)$ such that

$$N_r(x^*) \subset G_\alpha.$$

Choose a $n \in \mathbb{N}$ is large enough so that

$$2^{-n}\delta < r,$$

Then since $x^* \in I_n$, due to (2.6), we have

$$|x^* - y| \leq \frac{\delta}{2^n} < r \qquad \forall y \in I_n.$$

Thus

$$I_n \subset N_r(x^*) \subset G_\alpha.$$

It is contradiction to the fact that I_n is not covered by any finite subcollection of $\{G_\alpha : \alpha \in A\}$. □

Lemma 2.77. *Let $k \in \mathbb{N}$, $E \subset \mathbb{R}^k$, and $x \in \mathbb{R}^k$. Assume that there exists a constant $\varepsilon > 0$ such that*

$$|x - z| \geq \varepsilon \qquad \forall z \in E.$$

Then x is not a limit point of E.

Proof. Assume that x is a limit point of E. Then there exists a $z \in E$ such that $x \neq z$ and

$$|x - z| < \varepsilon.$$

However, this is contradiction to the assumption

$$|x - z| \geq \varepsilon \qquad \forall z \in E.$$

□

Lemma 2.78. *Let $k \in \mathbb{N}$, $E_1 \subset \mathbb{R}^k$, $E_2 \subset \mathbb{R}^k$, $E = E_1 \cup E_2$, and x is a limit point of E. Assume that E_1 is finite. Then x is a limit point of E_2.*

Proof. Since E_1 is finite, $E_1 - \{x\}$ is also finite set. Thus we can put

$$E_1 - \{x\} = \{e_1, e_2, \ldots, e_n\}$$

for some positive integer n. Since $e_i \neq x$, $|x - e_i| > 0$ for all $i = 1, \ldots, n$. Set

$$r_0 = \min\{|x - e_1|, |x - e_2|, \ldots, |x - e_n|\}.$$

Since x is a limit point for any $0 < r < r_0$, there exist $z \in E$ such that $x \neq z$ and

$$|x - z| < r.$$

Since $|x - z| < r_0 \leq |x - e_i|$, $z \neq e_i$ for all $i = 1, \ldots, n$. Thus $z \notin E_1$ and $z \in E_2$. This implies that x is a limit point of E_2. □

Exercise 2.79. *Let (X, d) be a metric space, $E \subset X$, and $x \in X$. Assume that there exists a positive constant r_0 such that for any $0 < r < r_0$, there exists a $z \in E$ such that $z \neq x$ and $|x - z| < r$. Show that x is a limit point of E.*

Theorem 2.80. *Let $k \in \mathbb{N}$, $E \subset \mathbb{R}^k$, and x be a limit point of E. Then there exists a sequence x_1, x_2, … such that*

$$x_i \neq x_j \qquad \forall i \neq j$$

$x_n \in E$, and

$$|x - x_n| < \frac{1}{n} \qquad \forall n \in \mathbb{N}.$$

Moreover, if the set

$$S := \{x_1, x_2, \ldots\}$$

has a limit point y, then

$$x = y.$$

In other words, x is the only limit point of S.

Proof. Since x is a limit point of E, there exists a $x_1 \neq x$ such that $x_1 \in E$ and

$$|x - x_1| < 1.$$

Since $x \neq x_1$, $|x - x_1| \neq 0$. Since x is a limit point of E, there exists $x_2 \neq x$ such that $x_2 \in E$,

$$|x - x_2| < \min\left\{\frac{1}{2}, |x - x_1|\right\} \leq \frac{1}{2}$$

Then since $|x - x_2| \neq |x - x_1|$, $x_1 \neq x_2$. Similarly, since x is a limit point of E, there exists $x_3 \neq x$ such that $x_3 \in E$,

$$|x - x_3| < \min\left\{\frac{1}{3}, |x - x_2|\right\} \leq \frac{1}{3}$$

Then since $|x - x_3| < |x - x_2| < |x - x_1|$,

$$x_3 \neq x_2 \quad \text{and} \quad x_3 \neq x_1.$$

Thus inductively, we can find a sequence x_1, x_2, x_3, ... such that

$$x_i \neq x_j \qquad \forall i \neq j,$$

$x_n \in E$, and

$$|x - x_n| < \frac{1}{n} \qquad \forall n \in \mathbb{N}.$$

Assume that the set

$$S := \{x_1, x_2, \ldots\}$$

has a limit point y and $x \neq y$. Since $x \neq y$, $|x - y| \neq 0$. Put $\varepsilon = |x - y|$. Since $\frac{1}{n}$ goes to 0 as $n \to \infty$, there exists a positive integer N such that for all $n \geq N$,

$$\frac{1}{n} < \frac{\varepsilon}{2}.$$

Thus by the triangle inequality, for all $n \geq N$

$$|y - x_n| \geq |x - y| - |x_n - x| \geq |x - y| - \frac{1}{n} > \frac{\varepsilon}{2}. \tag{2.7}$$

Set

$$S_1 = \{x_1, x_2, \ldots, x_{N-1}\}$$

and

$$S_2 = \{x_N, x_{N+1}, x_{N+2}, \ldots\}.$$

Since y is a limit point of S, $S = S_1 \cup S_2$, and S_1 is a finite set, y is a limit point of S_2 by Lemma 2.78. Due to (2.7),

$$|y - z| \geq \frac{\varepsilon}{2} \qquad \forall z \in S_2.$$

Thus by Lemma 2.77, S_2 has no limit point. But it is contradiction to the fact that y is a limit point of S_2. □

Exercise 2.81. *Let $k \in \mathbb{N}$ and $E \subset \mathbb{R}^k$. Assume that E is bounded. Show that there exists a k-cell I such that $E \subset I$.*

Exercise 2.82. *Let $k \in \mathbb{N}$ and $E \subset \mathbb{R}^k$. Assume that E is not bounded.*

(*i*) *Show that there exists a sequence x_1, x_2, ... such that $x_n \in E$ and*

$$|x_n| > n \qquad \forall n \in \mathbb{N}.$$

(*ii*) *Additionally, prove that the set*

$$S := \{x_n : n \in \mathbb{N}\}$$

is infinite.

Lemma 2.83. *Let $k \in \mathbb{N}$ and x_1, x_2, $\cdots$ be a sequence in $\mathbb{R}^k$. Assume that*

$$|x_n| > n \qquad \forall n \in \mathbb{N}. \tag{2.8}$$

Then the set

$$S := \{x_1, x_2, \ldots\}$$

has no limit point in $\mathbb{R}^k$.

Proof. We use the reduction to absurdity. Suppose that S has a limit point x in $\mathbb{R}^k$. Then $|x|$ is a constant and thus there exists a positive integer N such that

$$|x| < N. \tag{2.9}$$

By the triangle inequality and (2.8),

$$n - |x| < |x_n| - |x| \leq |x - x_n| \qquad \forall n \in \mathbb{N}. \tag{2.10}$$

Due to Theorem 2.34, every neighborhood of x has infinitely many points of S. Thus there exist infinitely many distinct positive integers n_1, n_2, ... such that

$$|x - x_{n_i}| < 1 \qquad \forall i \in \mathbb{N}.$$

Moreover, since the set $\{n_1, n_2, \ldots\}$ is infinite and distinct positive integers,

$$n_j > 2N \tag{2.11}$$

for infinitely many j. Choose a n_j satisfying (2.11). Then by (2.9) and (2.10), we have

$$N = 2N - N < n_j - |x| < |x - x_{n_j}| < 1.$$

It is contradiction since N is a positive integer. □

Theorem 2.84 (Heine-Borel Theorem)**.** *Let $k \in \mathbb{N}$ and $E \subset \mathbb{R}^k$. The following three properties are equivalent, i.e. one of the properties implies the other two properties:*

(*a*) *E is closed and bounded*
(*b*) *E is compact*
(*c*) *Every infinite subset of E has a limit point in E.*

Proof. First we prove $(a) \Rightarrow (b)$. Assume that E is closed and bounded. Since E is bounded, there exists a k-cell I such that $E \subset I$. By Theorem 2.76, I is compact. Since E is a closed subset of the compact set I, E is also compact by Theorem 2.65.

Next we prove $(b) \Rightarrow (c)$. However, this is proved in Theorem 2.72.

Finally it is sufficient to show $(c) \Rightarrow (a)$. First we show that E is bounded.

If E is not bounded, there exists a sequence x_1, x_2, ... such that $x_n \in E$, and

$$|x_n| > n \qquad \forall n \in \mathbb{N}.$$

Then by Lemma 2.83, the set

$$S_1 = \{x_1, x_2, \ldots\}$$

does not have a limit point and is infinite. It is contradiction to the statement (c). Next we show that E is closed. We use the reduction to absurdity again. Assume

that E is not closed. Then there exists a limit point $y \in \mathbb{R}^k$ of E such that $y \notin E$. By Theorem 2.80, there exist y_1, y_2, ... such that

$$y_i \neq y_j \qquad \forall i \neq j$$

$y_n \in E$, and

$$|y - y_n| < \frac{1}{n} \qquad \forall n \in \mathbb{N}.$$

Moreover, we know that y is the only limit point of the set

$$S_2 := \{y_1, y_2, \ldots\}$$

and S_2 is infinite. By (c), S_2 has a limit point $s \in E$. Thus we have

$$s = y.$$

However, it is contradiction since $y \notin E$ and $s \in E$. Therefore E is closed. □

Exercise 2.85. *Let $k \in \mathbb{N}$, $E \subset K \subset \mathbb{R}^k$. Assume that K is compact. Prove or disprove the following three statements:*

(*i*) *Every subsets of E has a limit point in E*
(*ii*) *Every subsets of E has a limit point in K*
(*iii*) *E is compact.*

Theorem 2.86 (Weierstrass Theorem)**.** *Let $k \in \mathbb{R}^k$ and $E \subset \mathbb{R}^k$. Assume that E is bounded and infinite. Then E has a limit point in $\mathbb{R}^k$.*

Proof. Since E is bounded, there exists a k-cell I such that

$$E \subset I.$$

By Theorem 2.76, I is compact. Since E is an infinite subset of I, E has a limit point in I by Theorem 2.84. □

3. Numerical sequences and series

Definition 3.1. *Let (X, d) be a metric space, $\{p_n\}$ be a sequence in X, and $p \in X$. We say that $\{p_n\}$* **converges to** *p iff for every $\varepsilon > 0$, there exists a positive integer N such that for all $n \geq N$, $d(p_n, p) < \varepsilon$. If p_n converges to p, then p is called a limit of the sequence $\{p_n\}$ and we write $p_n \to p$ as $n \to \infty$ or*

$$\lim_{n\to\infty} p_n = p.$$

If there is a p such that $\lim_{n\to\infty} p_n = p$, then we simply say that the sequence $\{p_n\}$ **converges**. *If p_n does not converge, it is said to* **diverge**. *Sometimes we say $\{p_n\}$ converges in X or $\{p_n\}$ diverges in X to emphasize the given metric space.*

Exercise 3.2. *Let $X_1 = \mathbb{R}$, $X_2 = (0, \infty)$, and $p_n = \frac{1}{n}$ for all $n \in \mathbb{N}$. Show that the sequence $\{p_n\}$ converges in X_1 but diverges in X_2.*

Exercise 3.3. *Let a be a nonnegative constant. Assume that for all $\varepsilon > 0$,*

$$a < \varepsilon.$$

Show that $a = 0$.

Theorem 3.4. *Let (X, d) be a metric space and $\{p_n\}$ be a sequence in X.*